

Salifort Motors:

Reducing Employee Turnover Through Data-Driven Actions



Identifying turnover drivers and enabling proactive employee retention

Presenter: Ai-Duyen, Pham

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EXECUTIVE SUMMARY

Employee Retention Analysis

Salifort Motors

EXECUTIVE TAKEAWAY

Turnover is driven by three interrelated patterns: imbalanced workload, not worthy compensation, and limited career progress.

Salifort Motors can reduce preventable turnover by enacting innovative regulations and activating early intervention before resignation simultaneously.

1. BUSINESS PROBLEM

- ❑ Analysis covered **11,991 employees across 10 departments**.
- ❑ **24% of employees left the company**, including critical high performers.



Leadership needs to understand the **key turnover signals** and **act before employees leave**.

2. KEY FINDINGS

- ❑ Employees managing **6–7 projects and working 11–13 hours per day** achieve high performance but dominantly quit their job.
- ❑ **Top performers** managing moderate workload yet **low paid** are **6x higher turnover**.
- ❑ Limited career upwards (only 3% promotion reported) triggers significant turnover in mid-tenure (3-6 years).
- ❑ A polarization in workload causes 100% turnover in overutilization and 66% turnover in underutilization, which needs reviewing and balancing.

3. SOLUTIONS

1. **Company-wide Retention Policies**
 2. **Predictive Risk Alerts**
 3. **Targeted HR Intervention**
- ❑ **Protect high performers** through clear workload and overtime limits.
 - ❑ **Strengthen** promotion, compensation, and **career pathways**.
 - ❑ **Re-engage underutilized employees** through mentoring and development opportunities.
 - ❑ Deploy the predictive model to generate employee risk alerts, supporting early HR intervention.

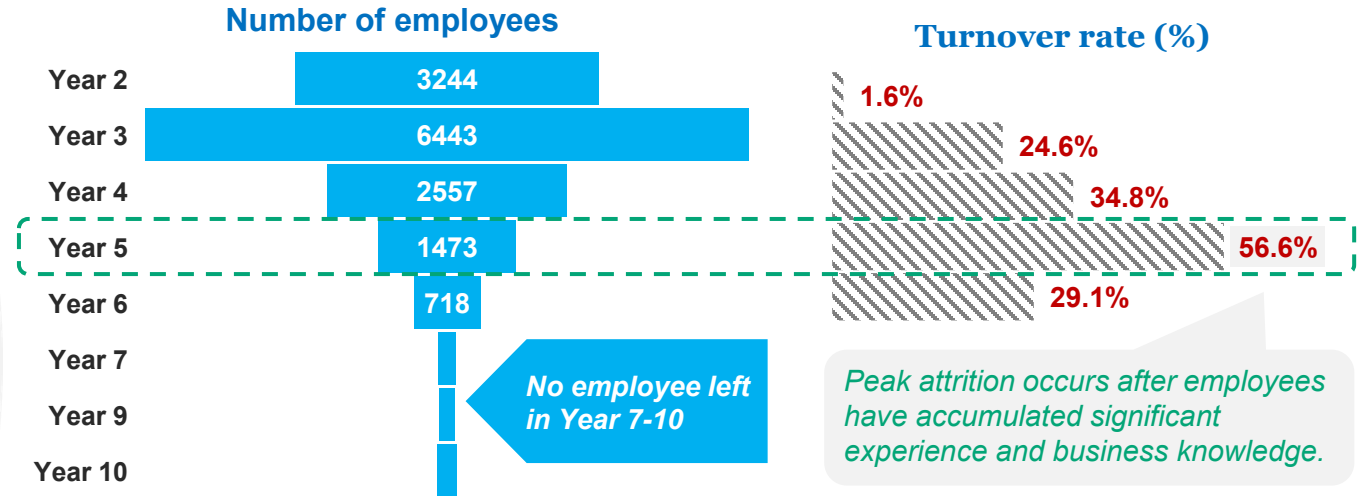
4. NEXT STEPS

- ❑ **Random Forest** was selected as **the champion model**, with **97.5% recall** and **93.0% precision** on the validation set.
- ❑ Employees should be grouped into **high-, medium-, and low-risk tiers**.
- ❑ High-risk alerts should **trigger confidential HR review**, not automated employment decisions.

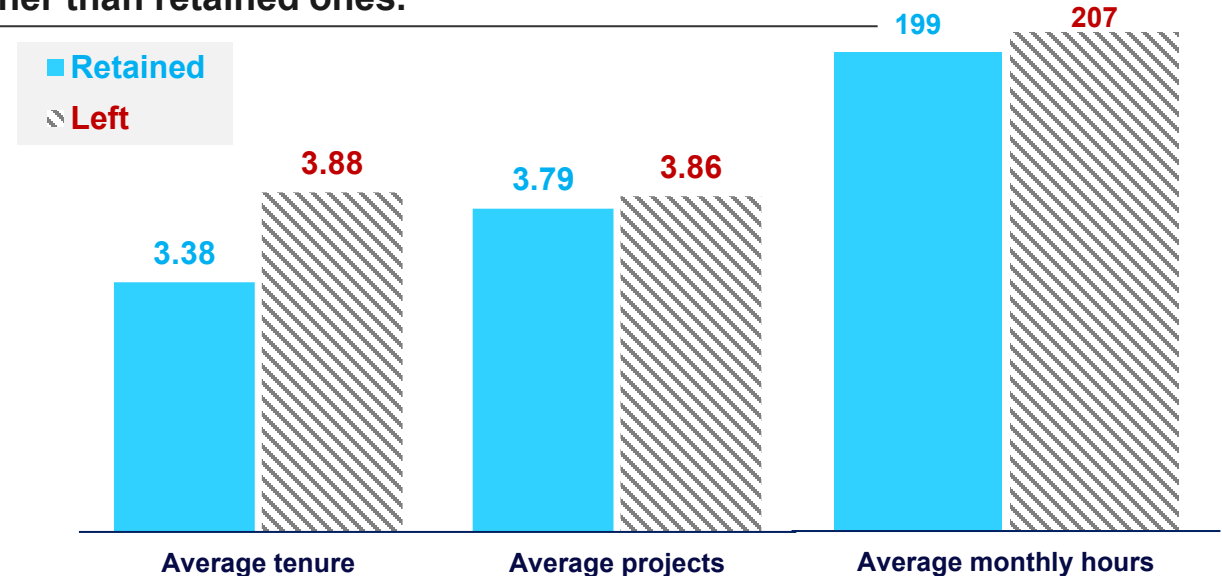
Employee turnover creates avoidable business and workforce risks

We've **lost 24% employees** over the last 5 years, many of whom contributed above the company average.

Mid-tenure attrition significantly erodes our investment in talents, particularly after they gain 3-5-year experience.

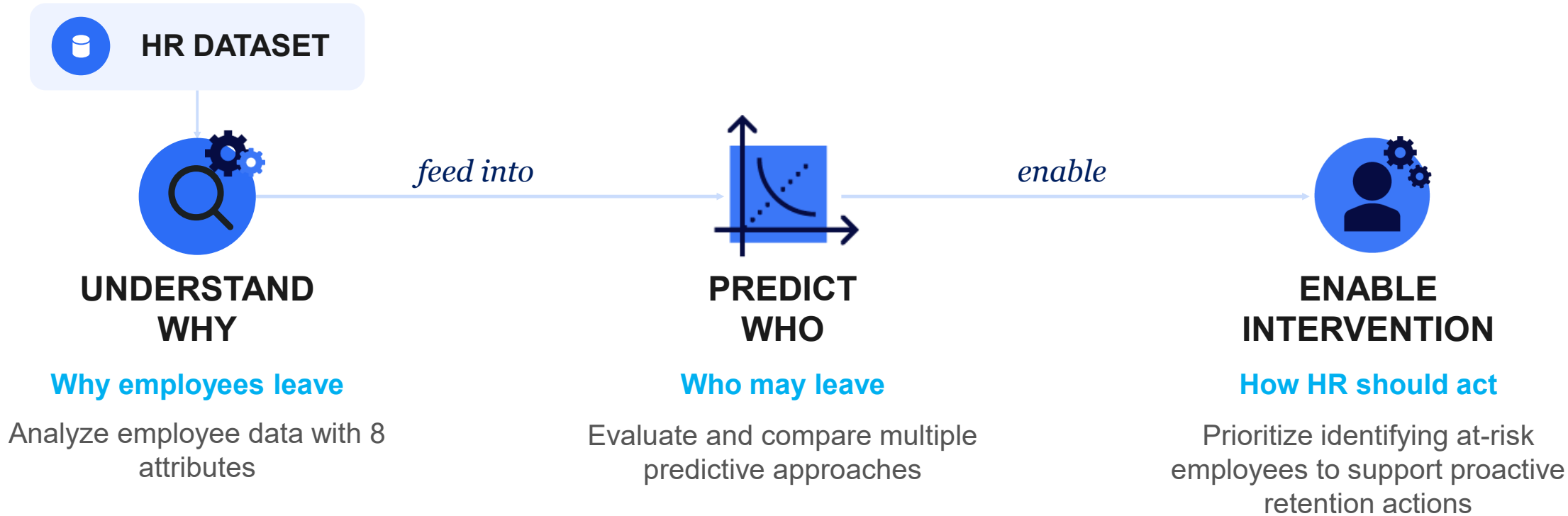


The average of tenure and contribution of employees who left are higher than retained ones.



The analysis moves HR from reactive turnover management to proactive prevention

Combining exploratory analysis and predictive modeling to enable targeted and proactive retention actions.



DELIVERABLES



Key turnover drivers



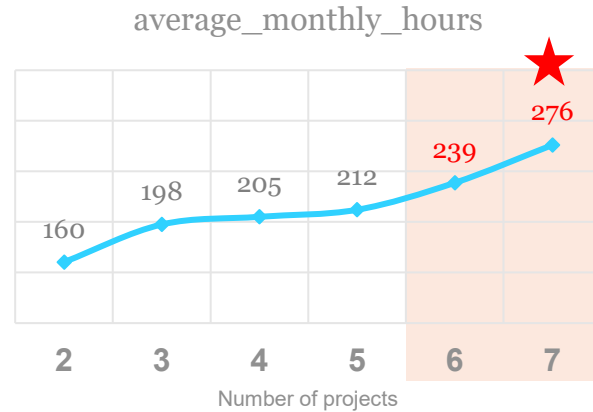
Retention policy recommendations



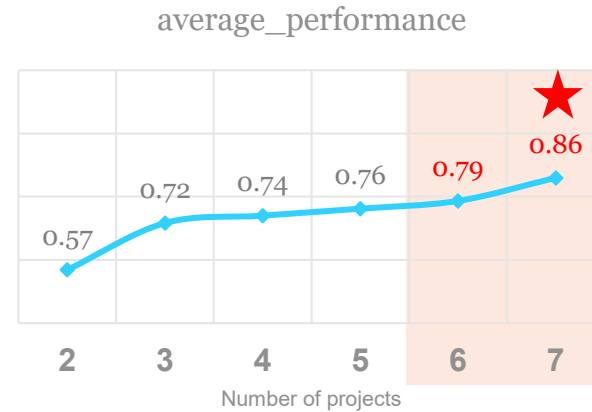
Early-warning intervention list

Finding 1: High performers with excessive workloads show poor satisfaction and likely to leave 3x than others

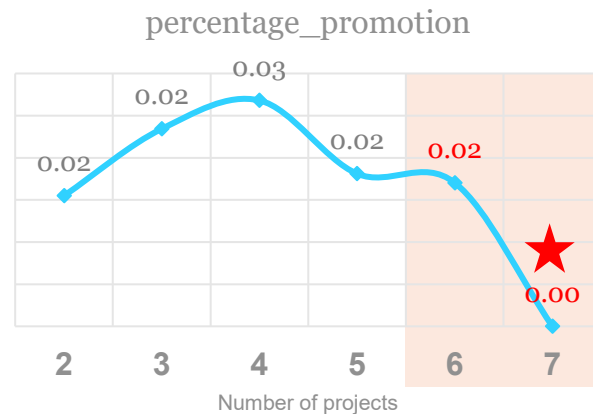
More projects leads to longer working hours...



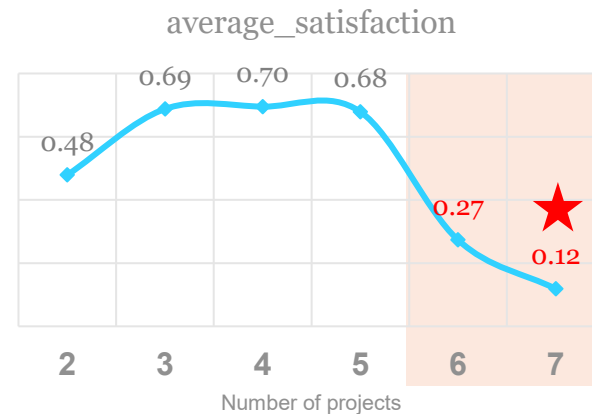
...achieved higher performance



But limited recognition...



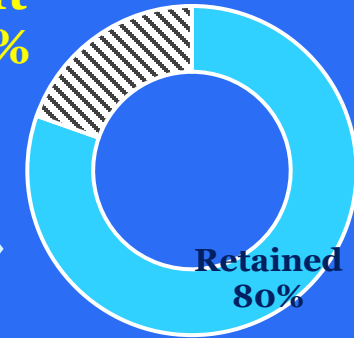
Then, they dissatisfied altogether.



Employees handling 6-7 projects are over 3X more likely to leave

Left 20%

Employees with 2-5 projects



Left 64%

Employees with 6-7 projects

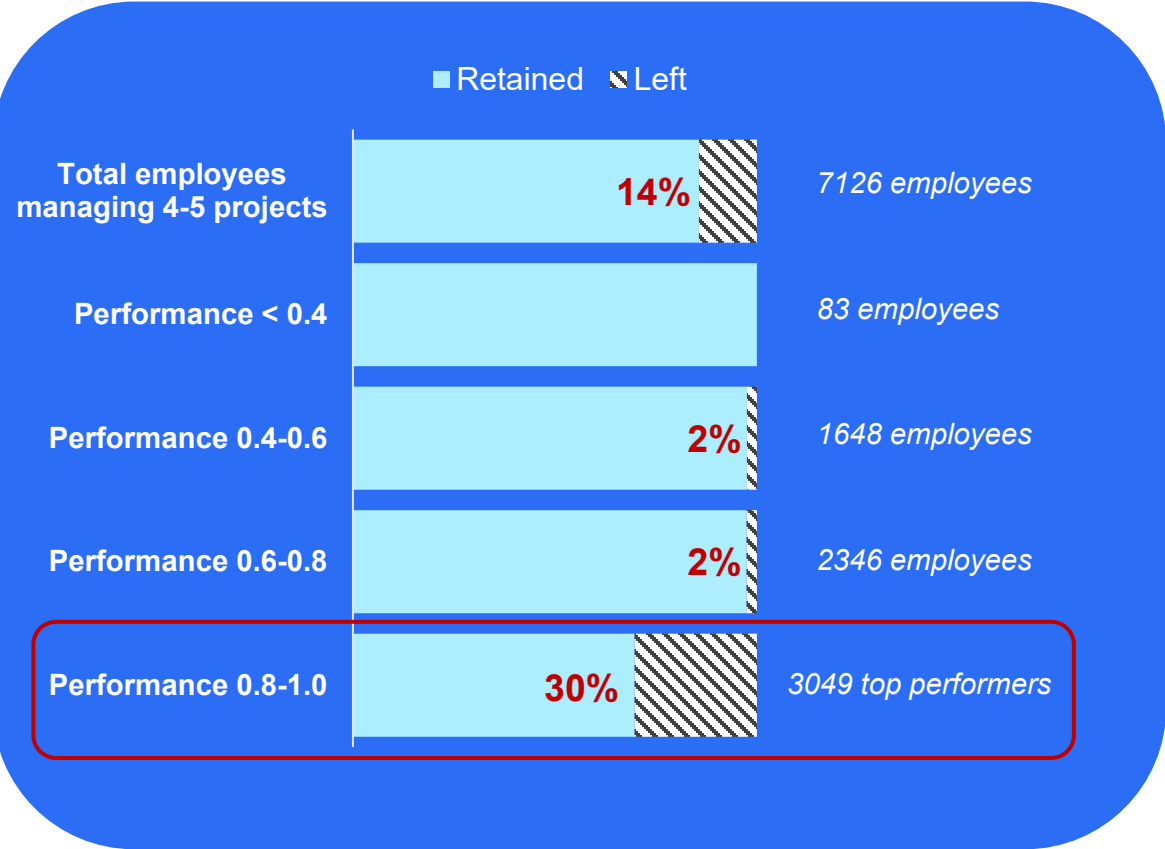


Business implication:

The company losing high-performers handling excessive workloads can be explained with their burnout or dissatisfied as workload growth is not matched by recognition and career progression.

Finding 2: Even satisfied top performers with 4-5 projects leave when compensation fall behind

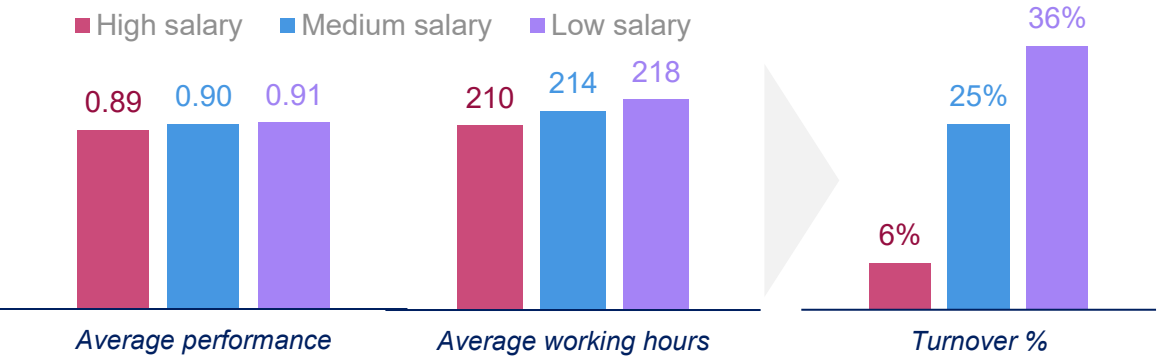
Among employees managing 4-5 projects, **30% top performers left**, more than twice the cohort average of 14%.



Most top performers who left were relatively high satisfaction



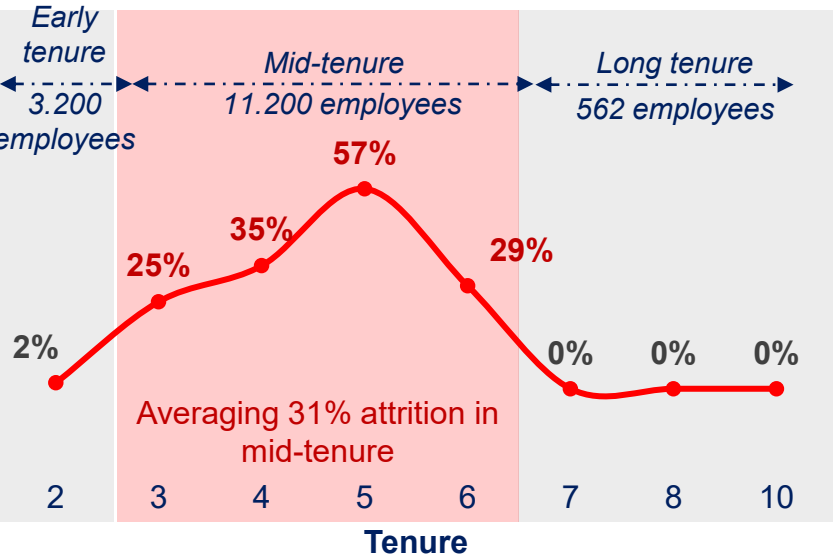
Comparable performance + Similar workload → **6x the turnover rate** for low-paid vs. high-paid top performers



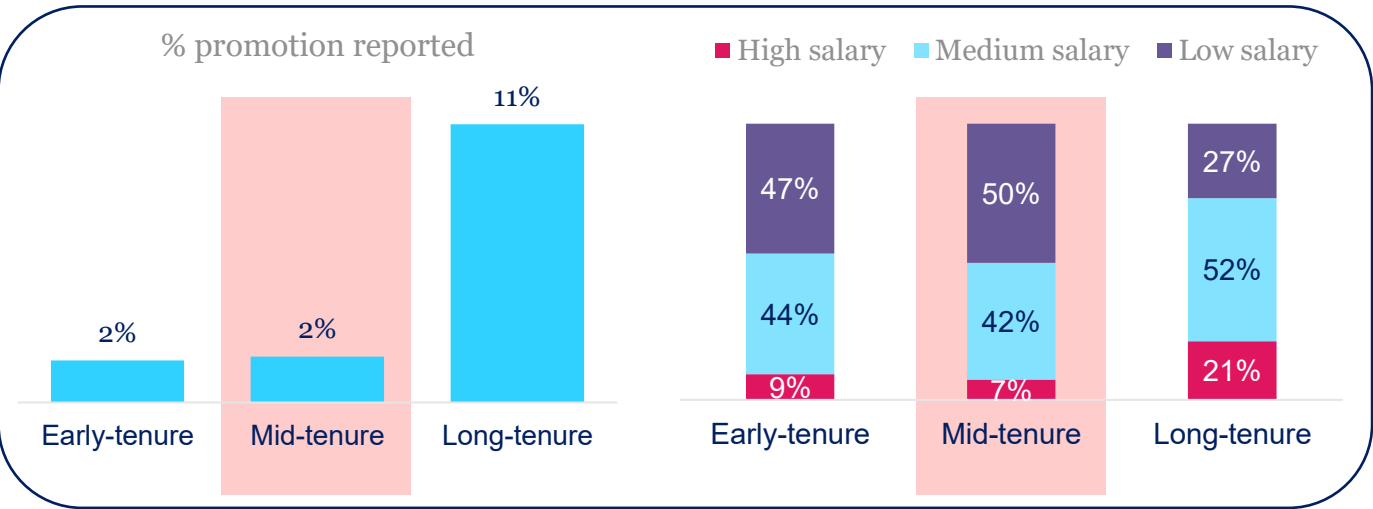
Business implication:
Satisfaction alone may not retain top performers, compensation should be investigated as potential retention gaps.

Finding 3: Turnover peaks during Years 3-6, when career progression becomes critical

Turnover rate of mid-tenure employees who have worked for 3-6 years is significantly higher than the average of 24%



Mid-tenure employees receive promotions at the same rate as freshers, while 50% remain in the low-salary group.



6% turnover

among employees who get a promotion in the last 5 years

24% turnover

among employees who **DO NOT** get a promotion in the last 5 years

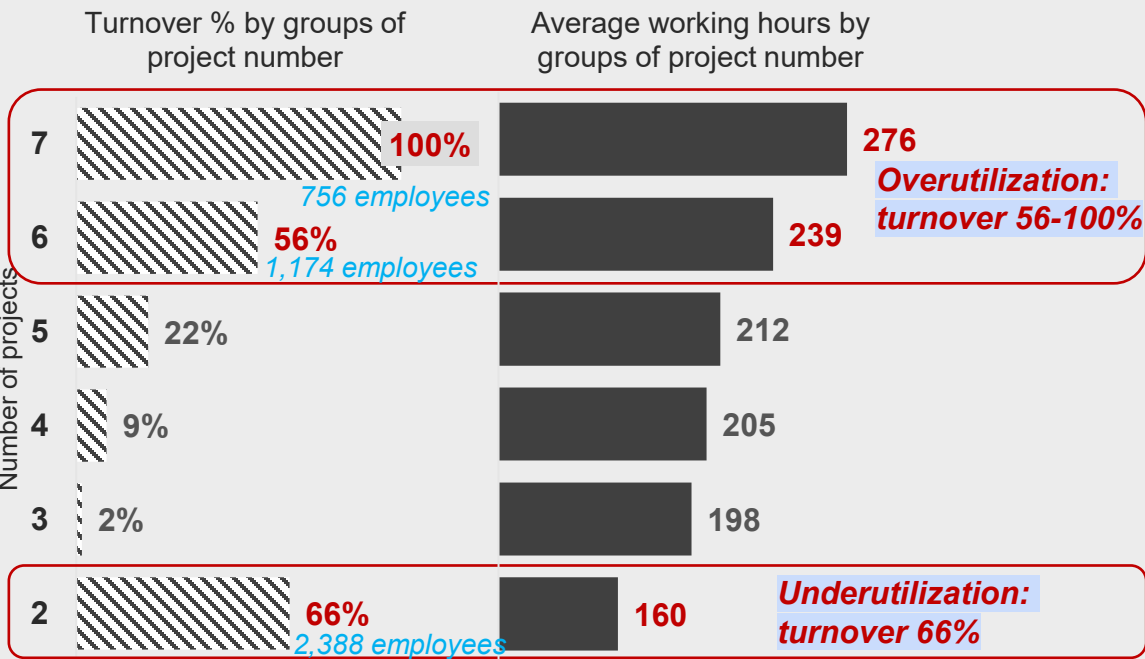
Promoted employees shows **75% lower attrition**

Business implication:

Waiting until long tenure to reward employees may be too late; career progression should occur during the critical Years 3-6 window.

Finding 4: Underutilized employees show 66% turnover, suggesting a potential gap in development opportunities

Turnover and average working hours across all employees:



Employees managing only two projects show 66% turnover, the second-highest rate across project-load groups. The same group records the lowest average working hours.

Among employees who left, those managing only two projects had lower working hours, performance, and tenure, indicating a potentially underdeveloped employee segment.

Employees who left by number of projects:

Group of left employees	Average working hours	Average performance (pts)	Average tenure (years)
2 projects	146 (< average)	0.51	3.02
3 projects	218 (> average)	0.75	3.95
4 projects	242 (> average)	0.88	4.90
5 projects	243 (> average)	0.90	5.03

The pattern may suggest limited development or assignment opportunities and should be investigated further.

Business implication:

Retention requires balanced utilization: protect overloaded employees while creating meaningful assignments and development opportunities for underutilized talent.

A holistic retention strategy combines targeted workforce policies with an early-warning system

Address company-wide attrition drivers

Pillar 1

1. Balance workload and protect high performers

- ❑ Establish clear overtime and workload guidelines.
- ❑ Review employees consistently working excessive hours or managing more than five projects.
- ❑ Recognize sustained additional effort through pay, time off, or workload adjustment.

2. Strengthen Career and Retention Pathways

- ❑ Begin structured career reviews from the third year of employment.
- ❑ Clarify promotion criteria and expand advancement opportunities.
- ❑ Align compensation and recognition with performance, tenure, and market value.

3. Develop Underutilized Employees

- ❑ Provide mentoring, training, and structured development plans.
- ❑ Assign employees to supporting roles on senior-led projects.
- ❑ Create a transparent path toward greater responsibility.

Pillar 2

Turn predictive insights into an early-warning system

Support HR department with early intervention by classifying employee risks into high, medium, and low risk. By regularly refreshed data, HR will be noticed with updated alert list.

Early alert: An early-warning model can identify employees at risk before resignation

Random Forest was selected because it identifies the highest proportion of actual leavers while keeping false alerts manageable.

97.5%



of actual leavers were correctly identified.
(outperforming 92.7% of XGBoost and 81.1% of Logistic Regression)

Only **7%**



of employees flagged as at risk were false alerts, helping keep HR review volume manageable.

Actionable early warning



- ✓ **Categorizes employees into high, medium and low risk**
- ✓ **Identifies the leading risk signals for each employee**
- ✓ **Directs HR toward the appropriate intervention**

The model is a prioritization tool for HR, not the intervention itself.

Appendix: Summary of model results and interpretation

- ❖ In analysis, 3 different models was trained and conducted to predict left employees, including Logistic Regression, Random Forest, and XGBoost.
- ❖ Recall is the North-star metric to choose the champion model, which means we choose a model that capture the most left employees.

Summary of model results:

Model	Precision	Recall	F1-score	Accuracy
Logistic Regression	0.57	0.81	0.67	0.86
Random Forest	0.93	0.97	0.95	0.98
XGBoost	0.96	0.92	0.94	0.98

- ❑ Both XGBoost and Random Forest show strong predictive ability, the Random Forest was preferred over XGBoost as higher recall at 97%.
- ❑ Logistic Regression remains useful as an interpretable baseline, but its lower performance suggests that it does not fully capture the dataset's non-linear patterns and feature interactions.

Top 5 important features:



Job satisfaction, workload, latest performance, and tenure are strong predictors in predicting the outcome target, which matches the key factors driving employees' attrition

**THANK YOU
FOR
YOUR
ATTENTION**

